For single phase concentric winding, we choose pole pitch as short pitch.

$$\text{Total no. of coils} = \frac{\textit{No of slots}}{2}$$

(Single layer)

$$= \frac{24}{2}$$

$$= 12$$

$$\text{No. of coils per pole} = \frac{\textit{No of coils}}{\textit{No of poles}}$$

(coils / pole)

$$= \frac{12}{4}$$

$$= 3$$

$$\text{No. of coils per phase} = \frac{\textit{No of coils}}{\textit{phases}}$$

(coils / phase)

$$= \frac{12}{1}$$

$$= 12$$

No. of coils per pole per phase

$$= \frac{\textit{No of coils}}{\textit{poles} \times \textit{phases}}$$

(coils / pole / phase)

$$= \frac{12}{4 \times 1}$$

$$= \frac{12}{4} = 3$$

Total no. of coil groups = poles × phases

$$= 4 \times 1 = 4$$

No. of coils per coil groups

$$= \frac{\textit{No of coils}}{\textit{No of coil groups}}$$

(coils / coil group)

$$= \frac{12}{4}$$

$$= 3$$

Pole 1	Pole 2	Pole 3	Pole 4
3	3	3	3

K - calculation (No of slots for 90°)

Electrical degree for a slot

(or)

Electrical degree between adjacent slots

$$\beta = \frac{P \times 180°}{S}$$

$$= \frac{4 \times 180°}{24}$$

$$= \frac{720°}{24}$$

$$= 30°$$

$$\text{No of slots for } 90° \text{ (K)} = \frac{90°}{30°} = 3$$

Hence,

Starting slot of the starting winding

$$= 1^{st} \text{ slot}$$

Starting slot of the running winding

$$= 1 + K$$

$$= 1 + 3$$

$$= 4^{th} \text{ slot}$$

So the starting winding must be inserted from the 1st slot and the running winding must be inserted from the 4th slot.

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